

Kidney Disease Prediction Using Machine Learning

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Kaggle link: <https://www.kaggle.com/datasets/amanik000/kidney-disease-dataset/code>

GitHub repo: <https://github.com/Anourine/kidney-disease-prediction>

Video link: https://nileuniversity-my.sharepoint.com/personal/a_wael2140_nu_edu_eg/_layouts/15/stream.aspx?id=%2Fpersonal%2Fa%5Fwael2140%5Fnu%5Fedu%5Feg%2FDocuments%2FRecording%2D20250606%5F222220%2Ewebm&nav=%7B%22defaultNavPanel%22%3A%7B%22pluginName%22%3A%22MediaSettingsLayer%22%7D%7D&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview%2Ee6239871%2D76da%2D4093%2D9c4d%2D9057673811b9

Introduction

Kidney disease is a progressive and often silent condition, typically going undiagnosed until it has reached a critical stage. Early detection is essential to reduce long-term complications and improve patient outcomes. While traditional diagnosis methods rely on lab tests and clinical expertise, the growth of healthcare data and advances in machine learning (ML) offer new opportunities to enhance disease prediction. This project applies machine learning to clinical data in order to detect kidney disease earlier and support healthcare decision-making with data-driven insights.

Objective

The objective of this project is to develop and evaluate machine learning models that can predict the likelihood of kidney disease in patients, based on clinical and biochemical indicators collected from routine check-ups and laboratory tests.

Dataset Description

The dataset used in this project contains over 20,000 patient records and approximately 43 features. These include:

- Vital signs (e.g., blood pressure)
- Biochemical metrics (e.g., serum creatinine, glucose levels)
- Other clinical values relevant to kidney function

The target variable is multiclass, indicating whether a patient has high risk of disease, severe risk, moderate risk, low risk and no disease

Methodology

This project follows a structured ML pipeline, mirroring the steps required in a real-world healthcare analytics system.

1-Data Collection

The dataset was gathered from real clinical records. It includes patient data such as blood pressure, creatinine, glucose, and other metrics widely used in assessing kidney function.

2. Data Preprocessing

Data preprocessing was critical to ensure consistency, accuracy, and compatibility with ML models:

- **Missing Value Imputation:** Missing values in numeric columns were filled using the mean; for categorical data, the most frequent value was used.
- **Label Encoding:** Categorical features were encoded into numerical values using label encoding to ensure compatibility with ML algorithms.
- **Outlier Removal:** The Z-score method was applied to identify and remove outliers (i.e., values more than 3 standard deviations away from the mean).
- **Feature Scaling:** StandardScaler was used to standardize numerical features so that they contribute equally to model training.

3. Handling Class Imbalance with SMOTE

The dataset showed imbalance between the classes and high imbalance in the no disease class. To address this, the SMOTE method was applied only on the training set. SMOTE creates synthetic examples of the minority class, ensuring the model receives balanced training and avoids bias toward the majority class.

4. Exploratory Data Analysis (EDA)

Exploratory analysis helped uncover data characteristics and informed feature selection:

- **Distribution Analysis:** Visual tools such as histograms and boxplots revealed the distribution of key features.
- **Correlation Heatmap:** Correlations among numerical features were visualized to detect multicollinearity and relationships with the target.
- **Target Balance Visualization:** Class distribution was plotted, confirming the need for resampling with SMOTE.

5. Model Selection and Training

Four classification models were chosen for training based on their effectiveness and interpretability:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier

Each model was trained using the resampled training set and evaluated using the holdout test set.

6. Hyperparameter Tuning

To improve model performance, GridSearchCV with 5-fold cross-validation was used. This process systematically tests multiple parameter combinations and selects the best configuration using the F1 weighted score as the primary metric.

7. Model Evaluation

The following metrics were used to evaluate model performance on the test set:

- **Accuracy:** Correct predictions as a percentage of all predictions
- **Precision:** Proportion of positive predictions that were actually correct
- **Recall (Sensitivity):** Proportion of actual positives correctly identified
- **F1 Score:** Harmonic mean of precision and recall
- **ROC-AUC Score:** Measures the model's ability to distinguish between classes across threshold levels

Both individual and combined ROC curves were plotted for visual comparison.

8. Feature Importance

The logistic regression model provided insight into which features most strongly influenced the model's decisions. A feature importance plot was created, helping to interpret model behavior and understand the most predictive clinical indicators.

9. Model Saving and Deployment Readiness

The best-performing model was saved using joblib, along with the feature list, for future deployment. This setup allows for consistent, real-time predictions when new patient data becomes available — a step toward integration into healthcare decision-support tools.

Conclusion

This project demonstrates how machine learning can be effectively applied to clinical data for early kidney disease prediction. The model pipeline includes balanced data handling, optimized training, and meaningful feature interpretation all critical elements in developing a reliable decision-support system for medical use.

